In-Class Worksheet
Introduction to R and Data Presentation

Name: ______________________________ **Date:** ______________

Course: H524 Introduction to Biostatistics
Topic: Introduction to R, Data Presentation, Numerical Summary Measures

1 Part 1: Getting Started with R

1.1 Exercise 1.1: Basic R Operations

Work with a partner to complete these basic R calculations:

```r
# 1. Use R as a calculator
3 * 2 + 4

# 2. Create a simple vector of systolic blood pressure readings
blood_pressure <- c(170, 193, 110, 135, 135, 115)
```

Discussion Questions:
- How do you create a vector in R?
- What does the `c()` function do?

2 Part 2: Data Types and Classification

2.1 Exercise 2.1: Identifying Data Types

For each variable below, classify it as:
- **Nominal** (categorical without order)
- **Ordinal** (categorical with order)
- **Discrete** (countable numbers)
- **Continuous** (measurable numbers)

Variable	Type	Reasoning
Blood type (A, B, AB, O)		
Pain level (none, mild, moderate, severe)		
Number of hospitalizations this year		
Body temperature (°F)		
Treatment group (control, low dose, high dose)		

2.2 Exercise 2.2: AI Enhancement Activity

Using AI Tools: Use GitHub Copilot or another AI assistant to help you:

1. Generate R code to create vectors for each data type above
2. Ask AI: "What's the best way to store categorical data in R?"

Verification Step: Test the AI-generated code and discuss with your partner whether the suggestions make sense.

3 Part 3: Descriptive Statistics

3.1 Exercise 3.1: Measures of Central Tendency

Using the blood pressure data: `blood_pressure <- c(170, 193, 110, 135, 135, 115)`

Complete the calculations both by hand and in R:

```
# Calculate mean
mean_bp <-

# Calculate median
median_bp <-

# Find the mode (most frequent value)
# Hint: Use table() function

# Order the data
ordered_bp <-
```

Hand Calculations:

- Mean: ______________ mmHg
- Median: ______________ mmHg
- Mode: ______________ mmHg

R Results:

- Mean: ______________ mmHg
- Median: ______________ mmHg
- Mode: ______________ mmHg

4 Part 4: Measures of Variability

4.1 Exercise 4.1: Spread of Data

```r
# Calculate range
max_bp <- max(blood_pressure)
min_bp <- min(blood_pressure)
range_bp <- max_bp - min_bp

# Calculate variance
var_bp <-

# Calculate standard deviation
sd_bp <-

# Calculate standard error of the mean
n <- length(blood_pressure)
sem_bp <- sd_bp / sqrt(n)
```

Fill in your results:

- Range: ________________ mmHg
- Variance: ________________ $mmHg^2$
- Standard Deviation: ________________ mmHg
- Standard Error of Mean: ________________ mmHg

Discussion: What do these measures tell us about the blood pressure data?

5 Part 5: AI-Enhanced Data Exploration

5.1 Exercise 5.1: Using AI for Data Summary

Prompt for AI: "Help me create a comprehensive summary of this blood pressure dataset: c(170, 193, 110, 135, 135, 115). Include all relevant descriptive statistics and suggest appropriate visualizations."

Your AI Interaction:

1. Copy the AI-generated code here:

```r
# Paste AI code here
```

2. **Verification Steps:**
   - Run the code and check if results match your hand calculations
   - Do the suggested visualizations make sense for this data?
   - Are there any errors or improvements needed?
3. **What you learned from AI:**
   - New R functions discovered: ____
   - Visualization suggestions: ____
   - Any surprising insights: ____

6 Wrap-up Discussion

Reflection Questions:

1. When might you prefer mean vs. median as a measure of central tendency?
2. How can AI tools help with statistical analysis while ensuring accuracy?
3. What's one new R function or concept you learned today?

Next Class Preview: We'll explore data visualization with histograms and boxplots, plus dive deeper into R programming basics.

7 AI Usage Documentation

Required for all students:

- AI tools used: ____________________
- Key prompts: ____________________
- Verification methods: ____________________
- Errors found and corrected: ____________________